

Introduction to Information Retrieval

What is Information Retrieval

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- Task of finding relevant information (document / book / product) that satisfies the user's information need

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- Task of finding relevant information (document / book / product) that satisfies the user's information need



User

What is Information Retrieval

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User

Information Need:
Book for Analytics newbie

What is Information Retrieval

- Task of finding relevant information (document / book / product) that satisfies the user's information need



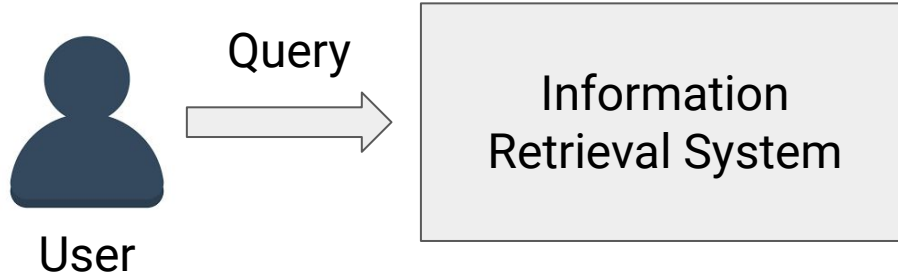
User

Information Need:
Book for Analytics newbie

Query

What is Information Retrieval

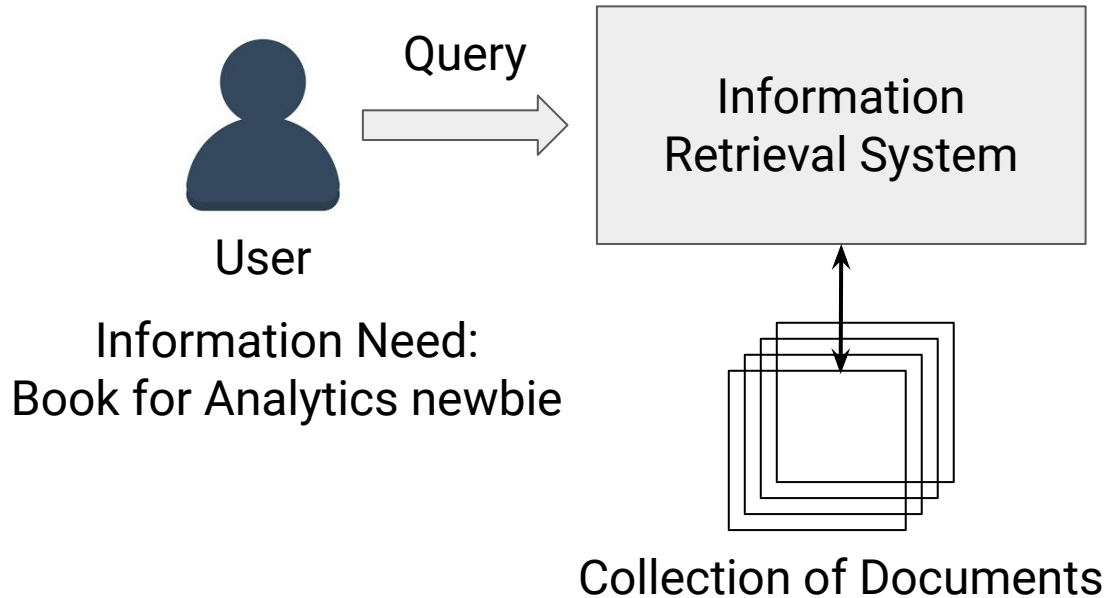
- Task of finding relevant information (document / book / product) that satisfies the user's information need



Information Need:
Book for Analytics newbie

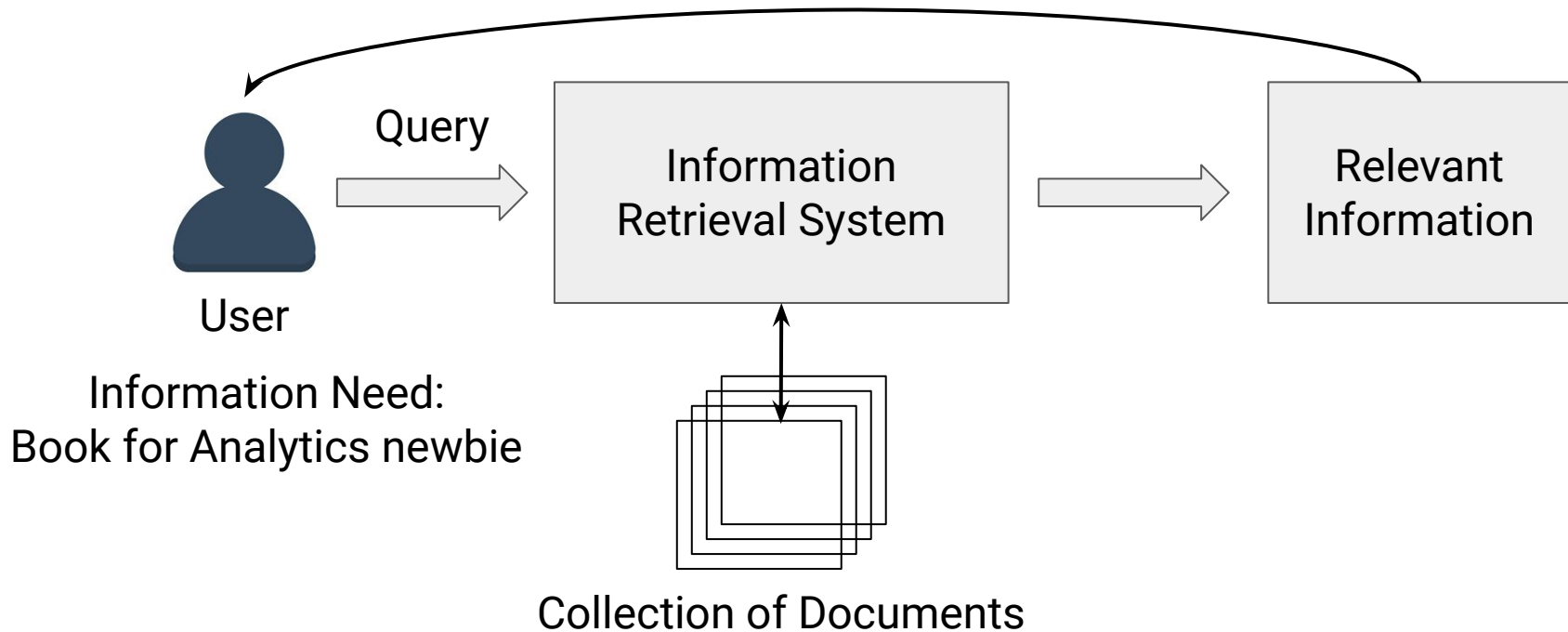
What is Information Retrieval

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Application of Information Retrieval

Application of Information Retrieval

- Library / Online book store

Application of Information Retrieval

- Library / Online book store
- Search Engines

Application of Information Retrieval

- Library / Online book store
- Search Engines
- E-Commerce stores

Approaches to Information Retrieval

Naive way to Information Retrieval

Naive way to Information Retrieval

Query: I want a book on Analytics

Naive way to Information Retrieval

Query: I want a book on Analytics

Collection: 4 documents related to either Analytics or Big-data or both

Naive way to Information Retrieval

Query: I want a book on Analytics

Collection: 4 documents related to either Analytics or Big-data or both

Processed query:

Documents containing the words Book and Analytics but not Big-Data

because the user dosnt mention big data

Naive way to Information Retrieval

Query: Book AND Analytics AND NOT Big-Data

Naive way to Information Retrieval

Query: Book AND Analytics AND NOT Big-Data

Solution:

- Scan all the documents

Naive way to Information Retrieval

Query: Book AND Analytics AND NOT Big-Data

Solution:

- Scan all the documents
- Find the documents containing the words Book AND Analytics

Naive way to Information Retrieval

Query: Book AND Analytics AND NOT Big-Data

Solution:

- Scan all the documents
- Find the documents containing the words Book AND Analytics
- Then remove the documents containing the word Big-Data

Challenge: Naive way to Information Retrieval

- Slow for large corpus of data

Term - Document Incidence Matrix

Query: Book AND Analytics AND NOT Big-Data

Term - Document Incidence Matrix

Term /
Words

Analytics				
Big-Data				
Book				
for				
The				
Tree				

Term - Document Incidence Matrix

Term / Words	Documents			
	Doc 1	Doc 2	Doc 3	Doc 4
	Analytics			
	Big-Data			
	Book			
	for			
	The			
	Tree			

Term - Document Incidence Matrix

Term / Words	Documents			
	Doc 1	Doc 2	Doc 3	Doc 4
	Analytics	1	1	0
	Big-Data			
	Book			
	for			
	The			
	Tree			

Term - Document Incidence Matrix

Term / Words	Documents				
		Doc 1	Doc 2	Doc 3	Doc 4
	Analytics	1	1	0	0
	Big-Data	0	1	0	0
	Book				
	for				
	The				
	Tree				

Term - Document Incidence Matrix

Term / Words	Documents				
		Doc 1	Doc 2	Doc 3	Doc 4
	Analytics	1	1	0	0
	Big-Data	0	1	0	0
	Book	1	1	1	0
	for	1	1	1	1
	The	1	1	1	1
	Tree	0	1	1	0

Term - Document Incidence Matrix

	Doc 1	Doc 2	Doc 3	Doc 4
Analytics	1	1	0	0
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Book	1	1	1	0
for	1	1	1	1
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Term - Document Incidence Matrix

Book

AND

Analytics

AND NOT

Big-Data

	Doc 1	Doc 2	Doc 3	Doc 4
Analytics	1	1	0	0
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Book	1	1	1	0
for	1	1	1	1
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Term - Document Incidence Matrix

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AND

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Big-Data

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Book	1	1	1	0
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The	1	1	1	1
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Term - Document Incidence Matrix

Book
AND
Analytics
AND NOT
Big-Data

	Doc 1	Doc 2	Doc 3	Doc 4
Analytics	1	1	0	0
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Book	1	1	1	0
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Term - Document Incidence Matrix

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AND

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AND NOT

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	Doc 1	Doc 2	Doc 3	Doc 4
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Book	1	1	1	0
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AND				

Term - Document Incidence Matrix

Book

AND

Analytics

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Big-Data

	Doc 1	Doc 2	Doc 3	Doc 4
Analytics	1	1	0	0
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Book	1	1	1	0
for	1	1	1	1
The	1	1	1	1
Tree	0	1	1	0
AND	1			

Term - Document Incidence Matrix

Book

AND

Analytics

AND NOT

Big-Data

	Doc 1	Doc 2	Doc 3	Doc 4
Analytics	1	1	0	0
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Book	1	1	1	0
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AND	1	0	0	0

Term - Document Incidence Matrix

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Term - Document Incidence Matrix

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Analytics

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Term - Document Incidence Matrix

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AND

Tree

AND NOT

Analytics

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Term - Document Incidence Matrix

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AND				

Term - Document Incidence Matrix

Book

AND

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Analytics

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Term - Document Incidence Matrix

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Book	1	1	1	0
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AND	0	0	1	0

Challenges: Term - Document Incidence Matrix

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- Inefficient with bigger collection

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- Inefficient with bigger collection
 - Number of documents: 1 million = 10^6

Challenges: Term - Document Incidence Matrix

- Inefficient with bigger collection
 - Number of documents: 1 million = 10^6
 - Number of unique words: 5,000

Challenges: Term - Document Incidence Matrix

- Inefficient with bigger collection
 - Number of documents: 1 million = 10^6
 - Number of unique words: 5,000
 - Size of incidence matrix = $5,000 \times 10^6$

Inverted Index

Inverted Index

- It consists of

Inverted Index

- It consists of
 - Dictionary of terms (vocabulary)

Inverted Index

- It consists of
 - Dictionary of terms (vocabulary)
 - Posting list for each term

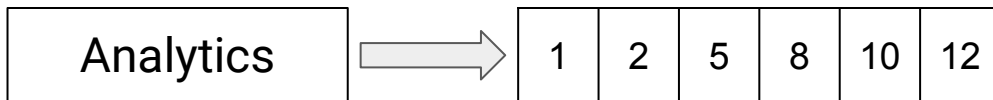
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Analytics

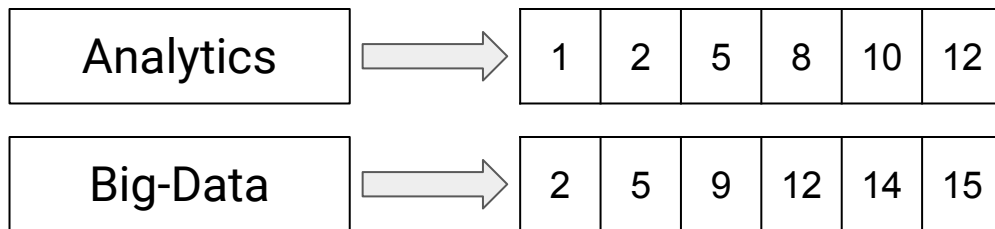
Inverted Index

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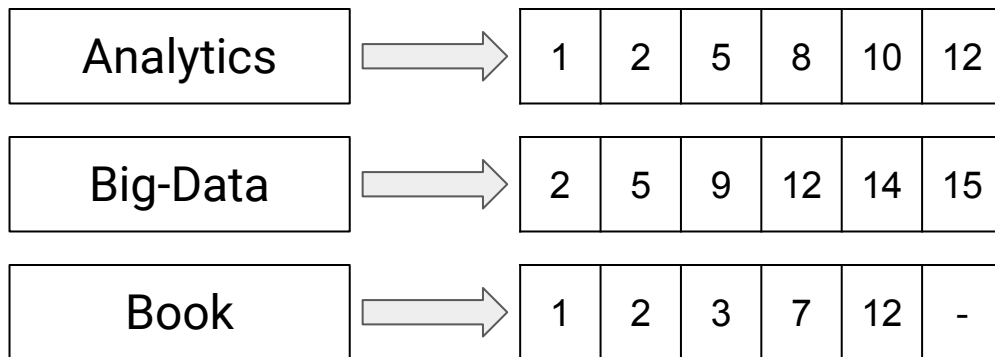
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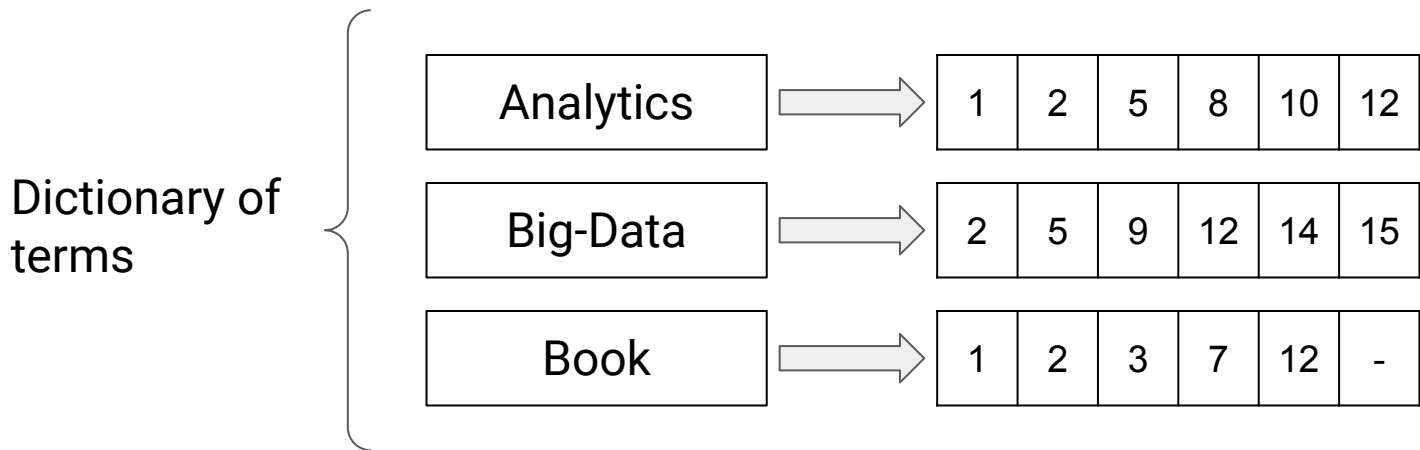
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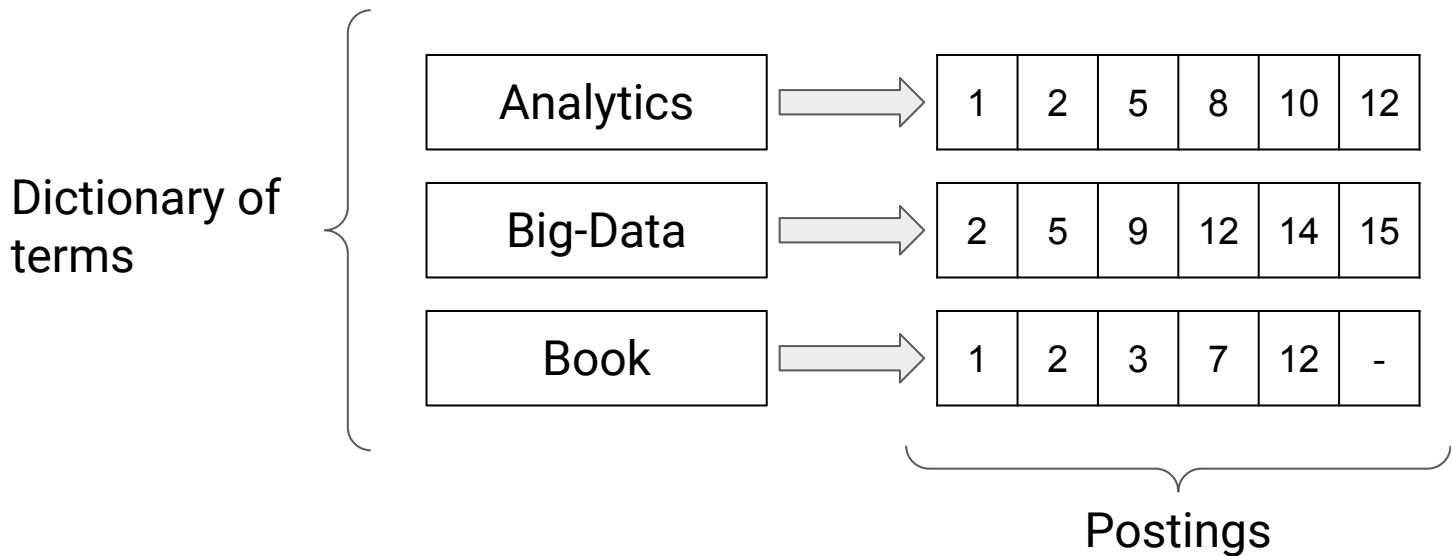
Inverted Index

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Inverted Index

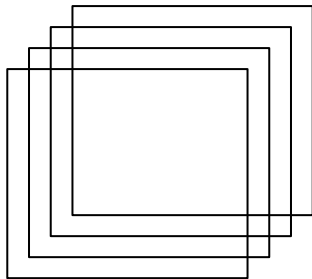
- It consists of
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 - Posting list for each term



Steps to create Inverted Index

Steps to create Inverted Index

1. Collect all the documents that needs to be indexed



This book is on Analytics

Steps to create Inverted Index

1. Collect all the documents that needs to be indexed
2. Perform tokenization

This book is on Analytics

Steps to create Inverted Index

1. Collect all the documents that needs to be indexed
2. Perform tokenization
3. Modify tokens

this book is on analytics

Steps to create Inverted Index

1. Collect all the documents that needs to be indexed
2. Perform tokenization
3. Modify tokens
4. Create inverted index using indexer

analytics

1	2	5	8	10	12
---	---	---	---	----	----

book

1	2	3	7	12	-
---	---	---	---	----	---

Steps of Indexer

Steps of Indexer

Doc 1

This book is on
analytics

Steps of Indexer

Doc 1

This book is on
analytics

Doc 2

Big-Data analytics
is the process of
examining large
volume of data

Steps of Indexer

1. Create token sequence

Doc 1

This book is on
analytics

Doc 2

Big-Data analytics
is the process of
examining large
volume of data

Term	DocID
This	1
book	1
is	1
on	1
analytics	1
Big-Data	2
analytics	2
is	2
the	2
process	2
of	2
examining	2
large	2
volume	2
of	2
data	2

Steps of Indexer

1. Create token sequence
2. Sort by terms
(And then on Doc ID)

Term	DocID		Term	DocID
This	1		analytics	1
book	1		analytics	2
is	1		Big-Data	2
on	1		book	1
analytics	1		data	2
Big-Data	2		examining	2
analytics	2	→	is	1
is	2		is	2
the	2		large	2
process	2		of	2
of	2		of	2
examining	2		on	1
large	2		process	2
volume	2		the	2
of	2		This	1
data	2		volume	2

Steps of Indexer

1. Create token sequence
2. Sort by terms
(And then on Doc ID)
3. Create dictionary of terms
and postings

Term	DocID
analytics	1
analytics	2
Big-Data	2
book	1
data	2
examining	2
is	1
is	2
large	2
of	2
of	2
on	1
process	2
the	2
This	1
volume	2

Steps of Indexer

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is	1
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of	2
on	1
process	2
the	2
This	1
volume	2

analytics
Big-Data
book
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Steps of Indexer

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of	2
of	2
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the	2
This	1
volume	2

analytics	1	2
Big-Data	2	
book	1	
data	2	
examining	2	
is	1	2
large	2	
of	2	
on	1	
process	2	
the	2	
this	1	
volume	2	

Processing queries using Inverted Index

Processing queries using Inverted Index

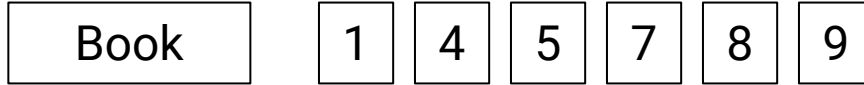
Book

AND

Analytics

Processing queries using Inverted Index

Book

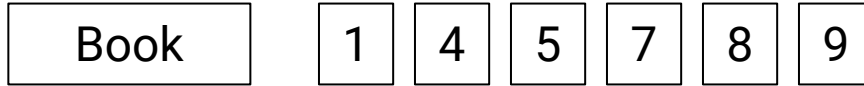


AND

Analytics

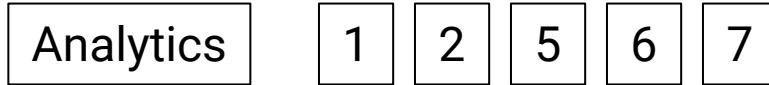
Processing queries using Inverted Index

Book

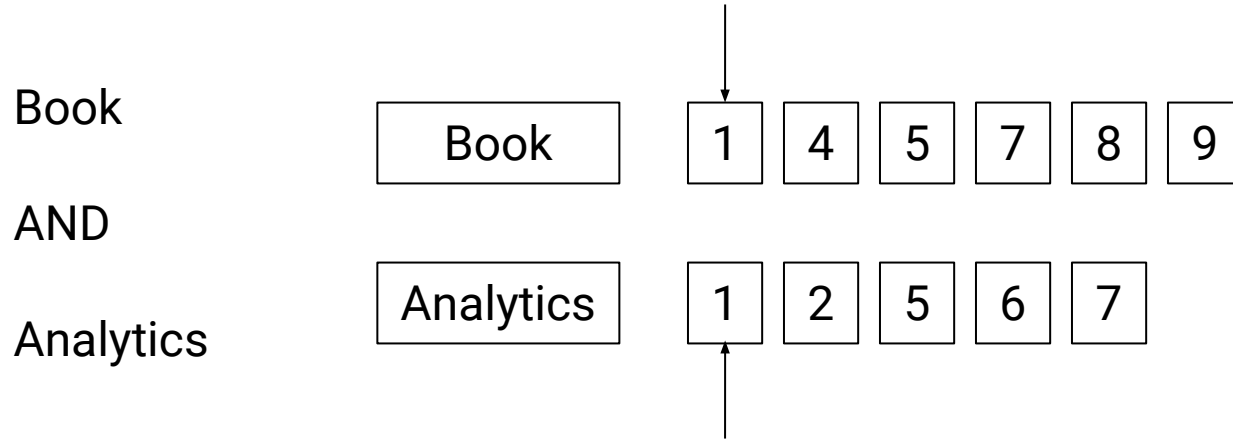


AND

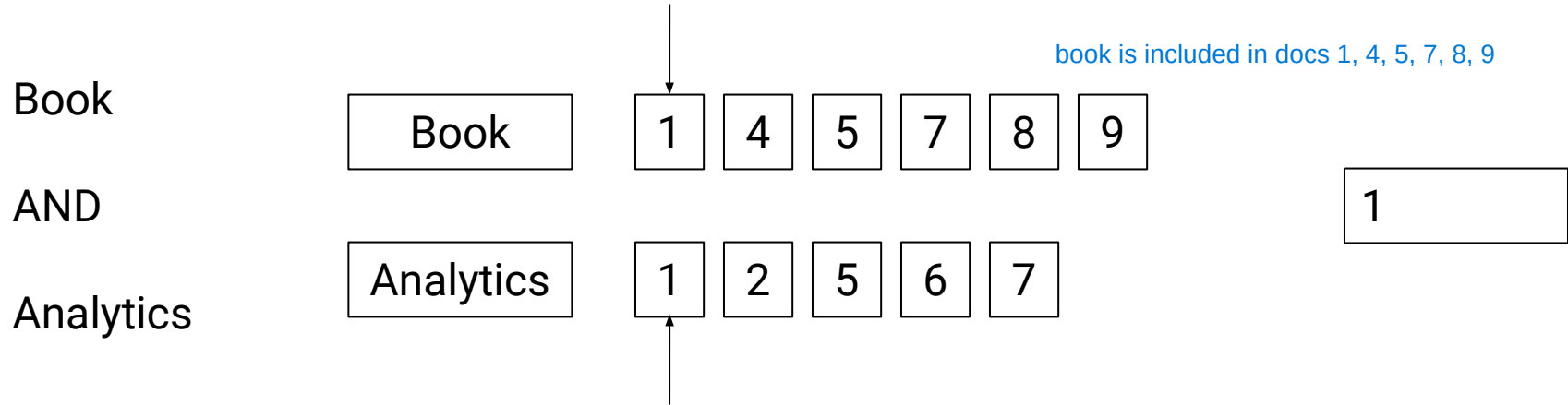
Analytics



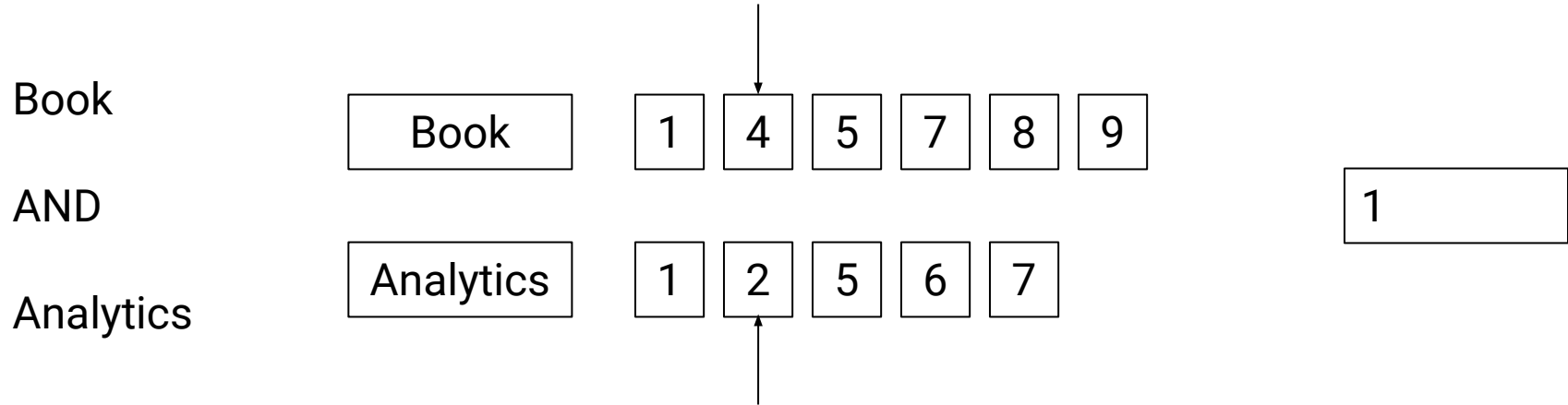
Processing queries using Inverted Index



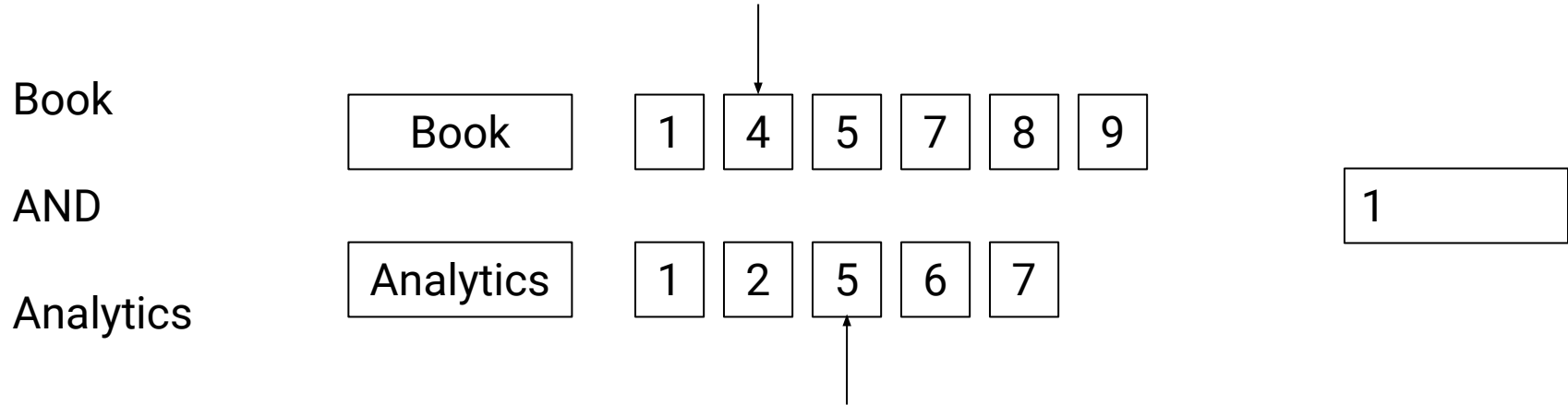
Processing queries using Inverted Index



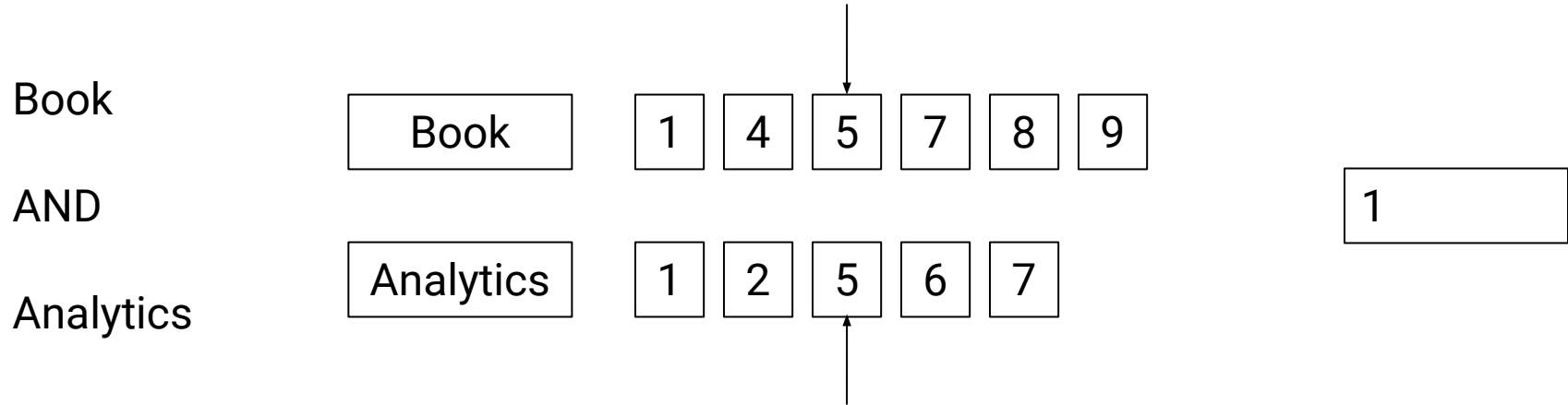
Processing queries using Inverted Index



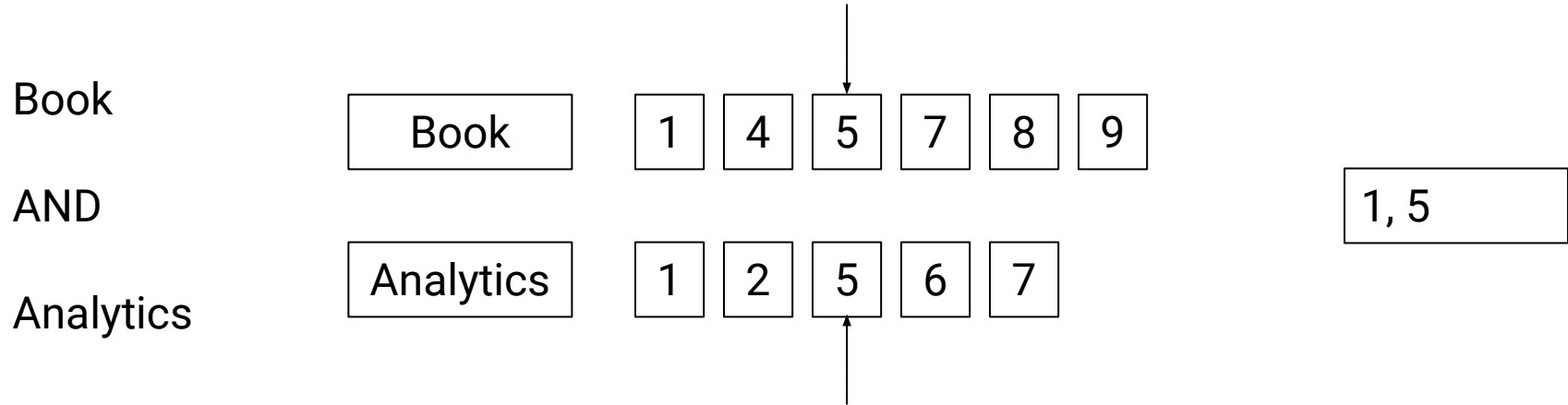
Processing queries using Inverted Index



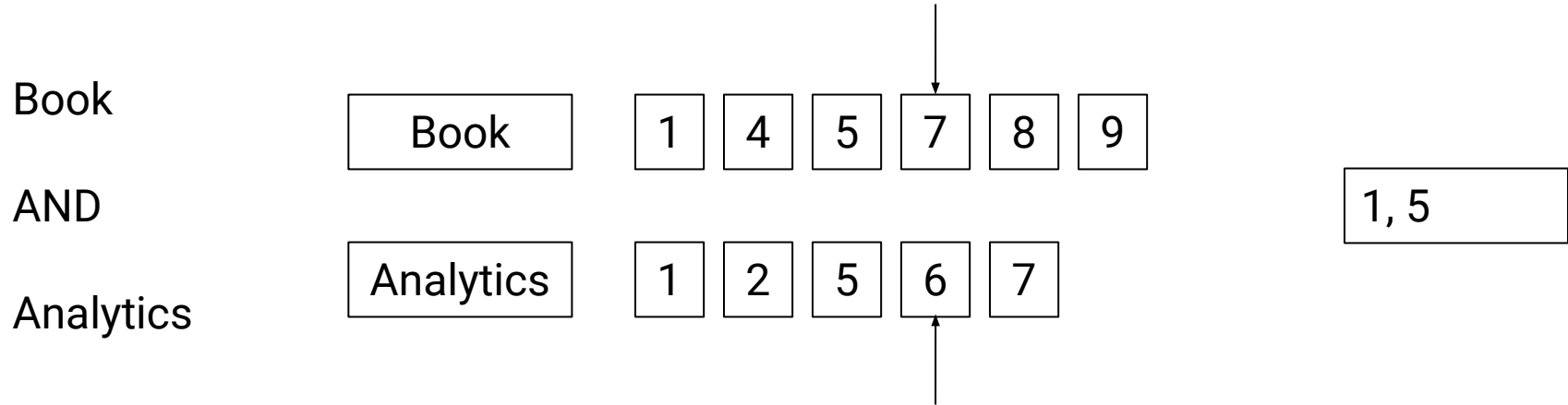
Processing queries using Inverted Index



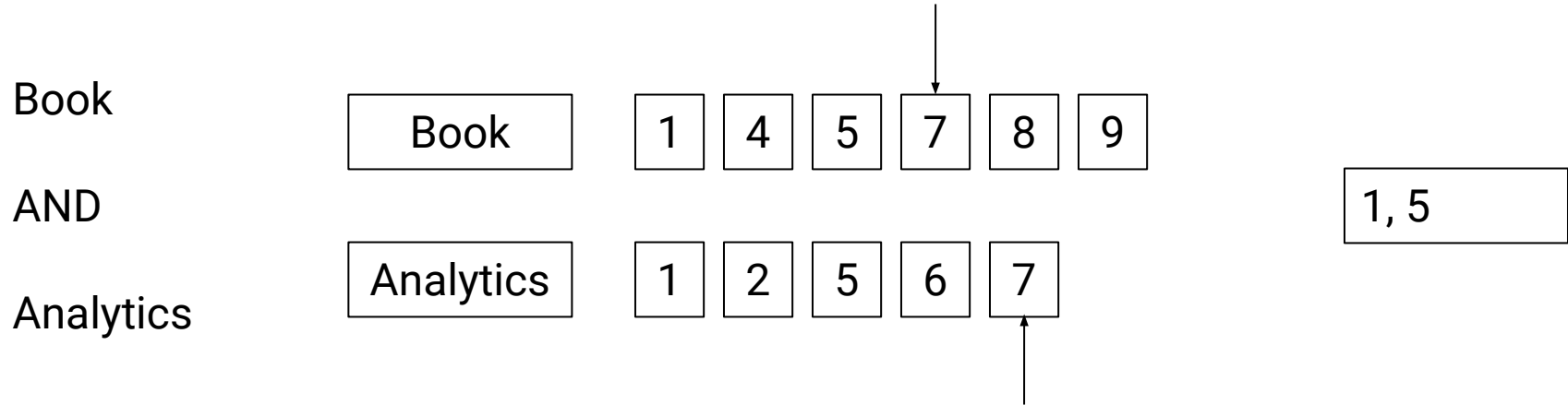
Processing queries using Inverted Index



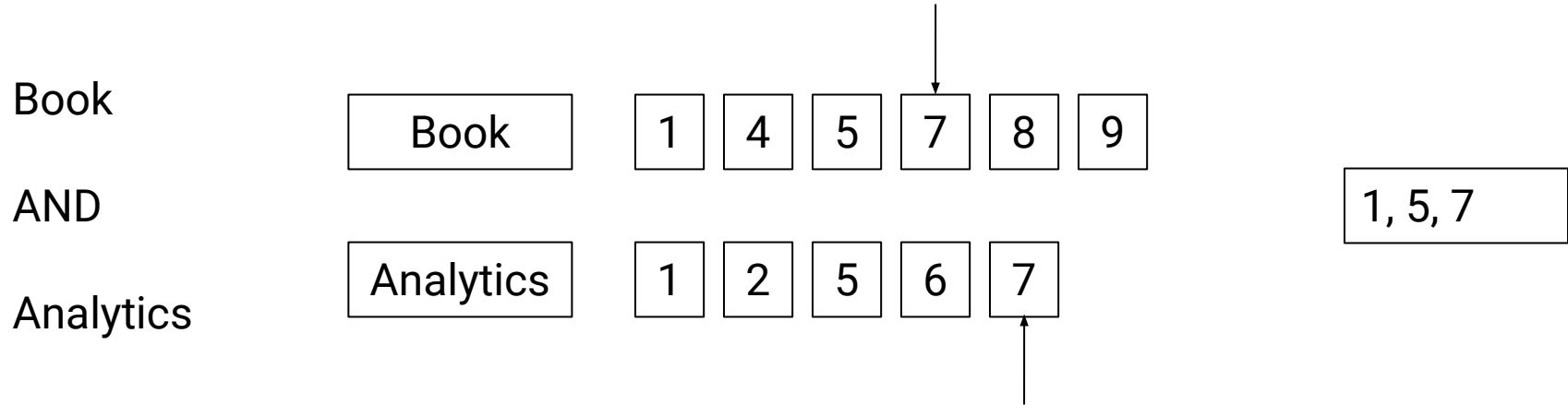
Processing queries using Inverted Index



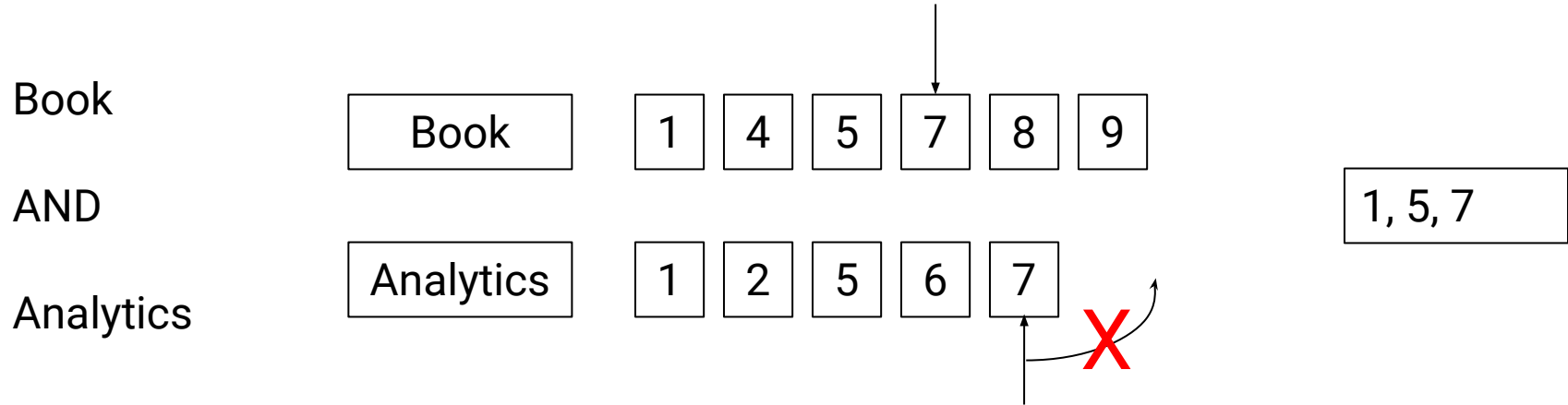
Processing queries using Inverted Index



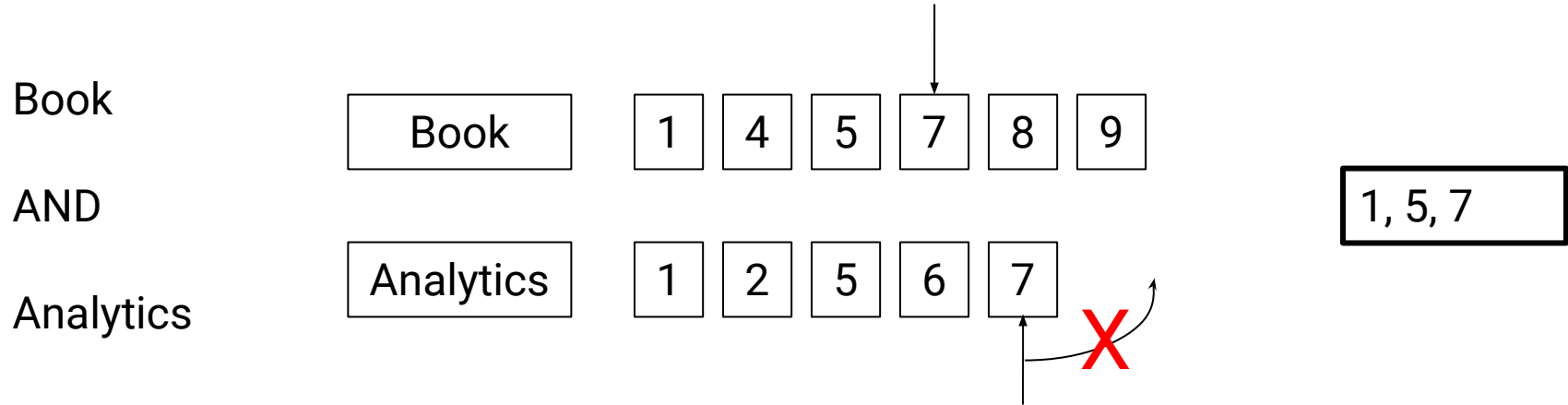
Processing queries using Inverted Index



Processing queries using Inverted Index



Processing queries using Inverted Index



Evaluation Metrics for Information Retrieval

Goal: Information Retrieval

- Retrieve documents with information that is relevant to user's information need

Goal: Information Retrieval

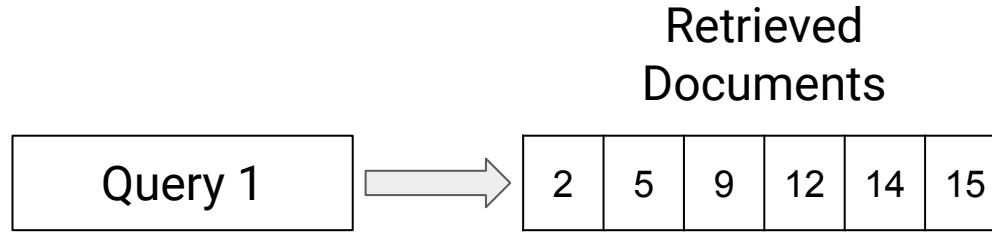
- Retrieve documents with information that is relevant to user's information need
- Given a query:
 - Return list of relevant documents

How good are the retrieved documents?

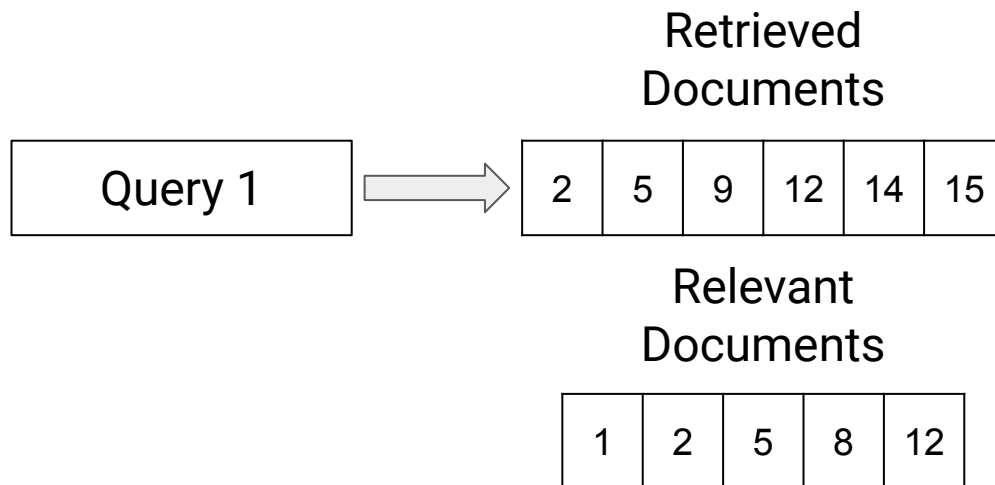
How good are the retrieved documents?

Query 1

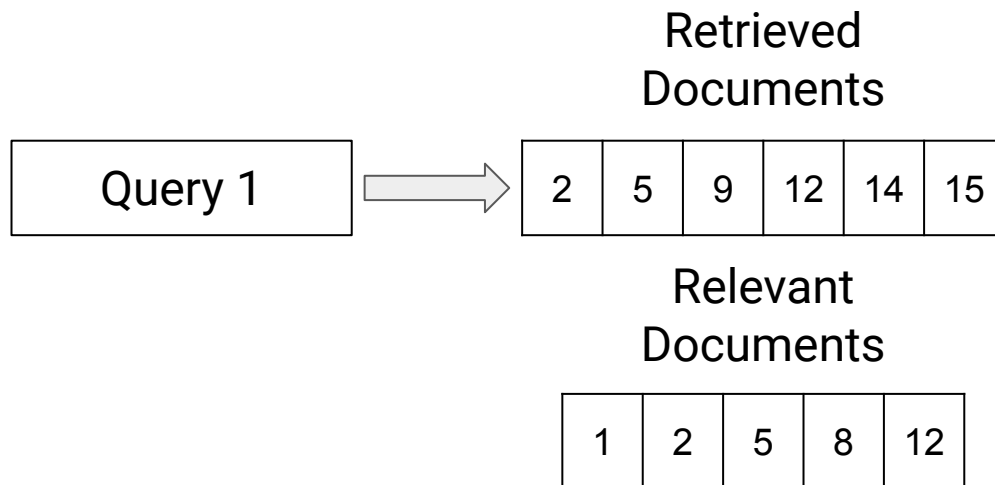
How good are the retrieved documents?



How good are the retrieved documents?

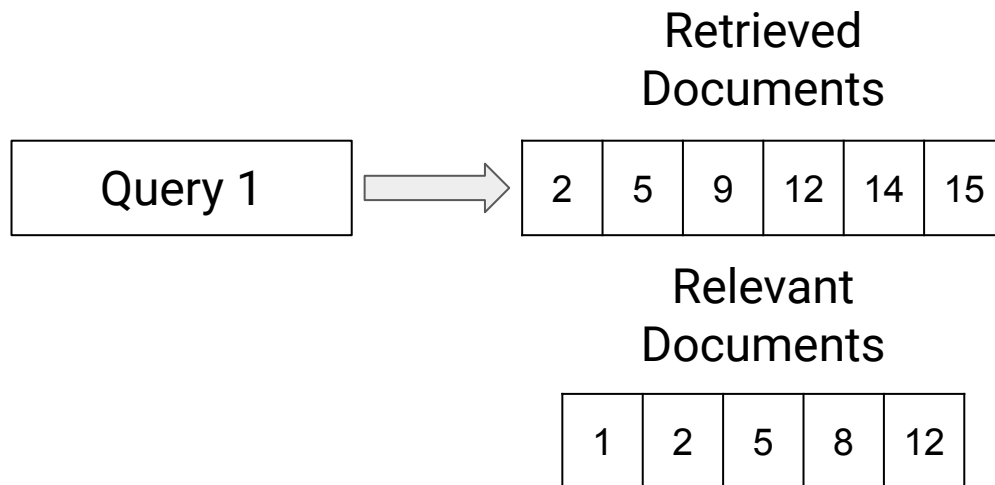


How good are the retrieved documents?



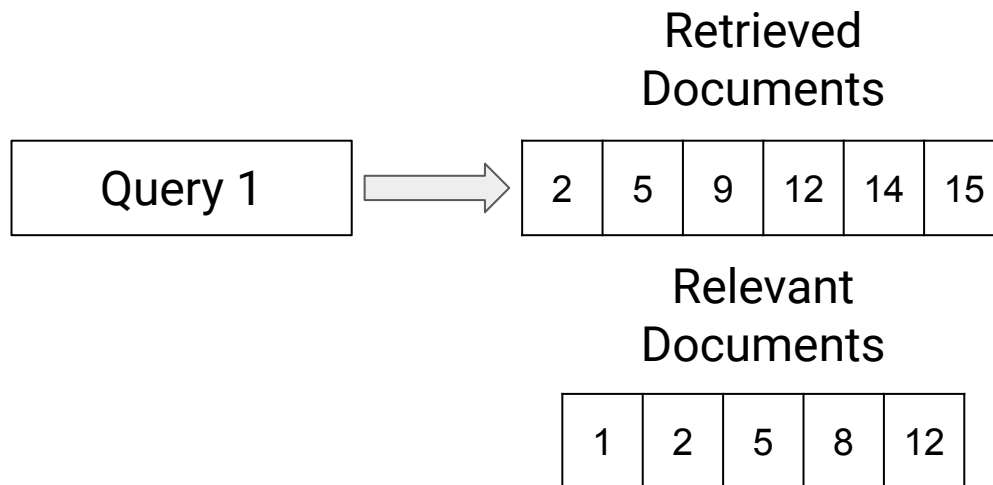
- Precision

How good are the retrieved documents?



- Precision: Fraction of retrieved documents, relevant to user's need

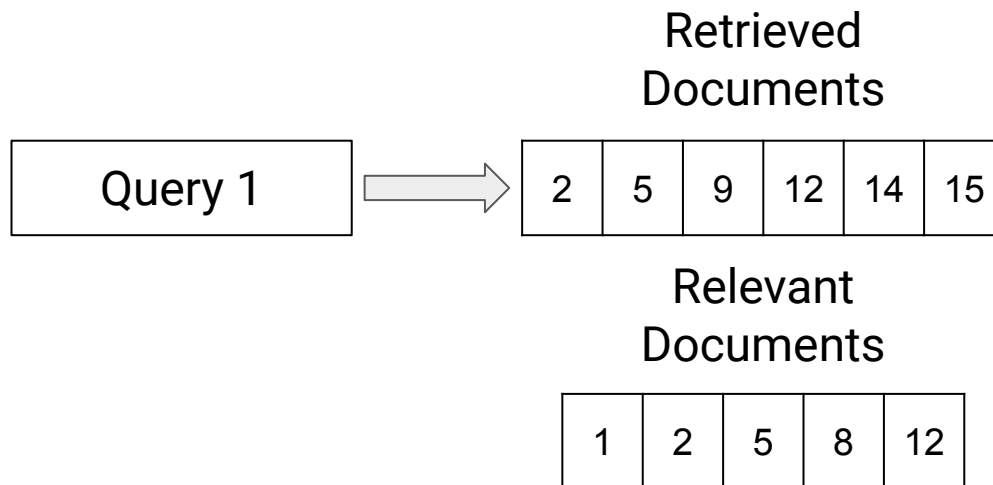
How good are the retrieved documents?



- Precision: Fraction of retrieved documents, relevant to user's need

Precision =

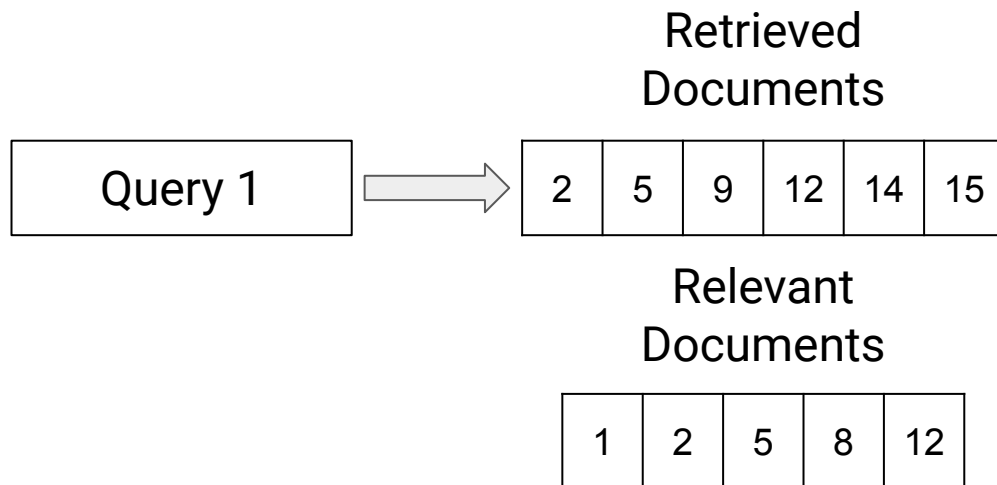
How good are the retrieved documents?



- Precision: Fraction of retrieved documents, relevant to user's need

Precision = $\frac{\text{Number of relevant documents retrieved}}{\text{Number of retrieved documents}}$

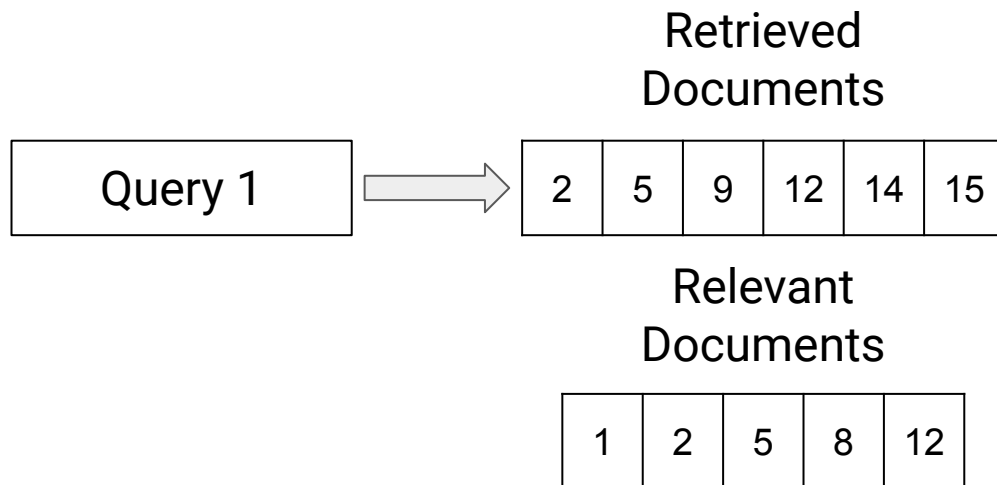
How good are the retrieved documents?



- Precision: Fraction of retrieved documents, relevant to user's need

$$\text{Precision} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of retrieved documents}}$$

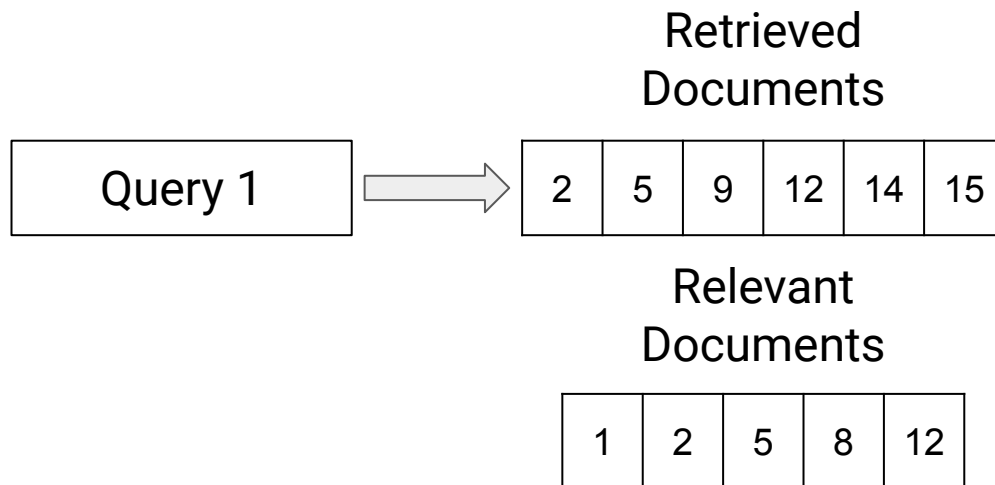
How good are the retrieved documents?



- Precision: Fraction of retrieved documents, relevant to user's need

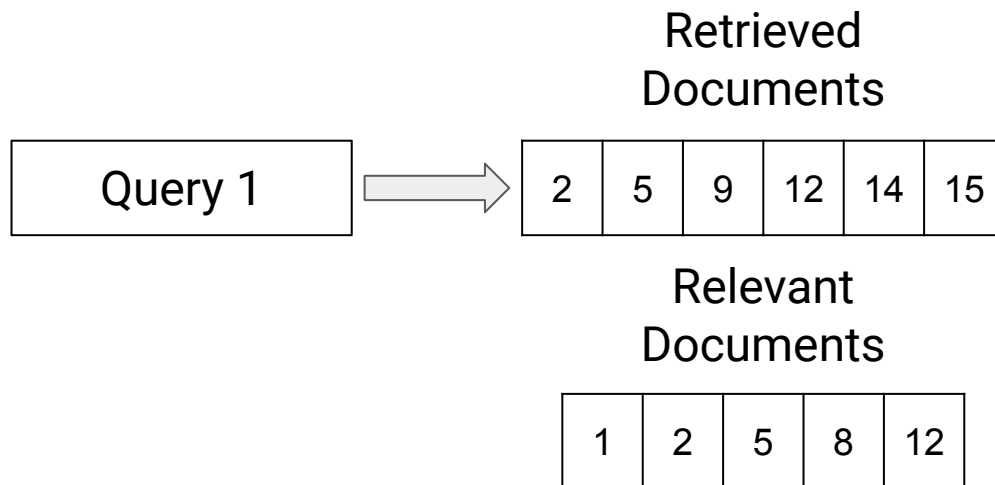
$$\text{Precision} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of retrieved documents}} = \frac{3}{6}$$

How good are the retrieved documents?



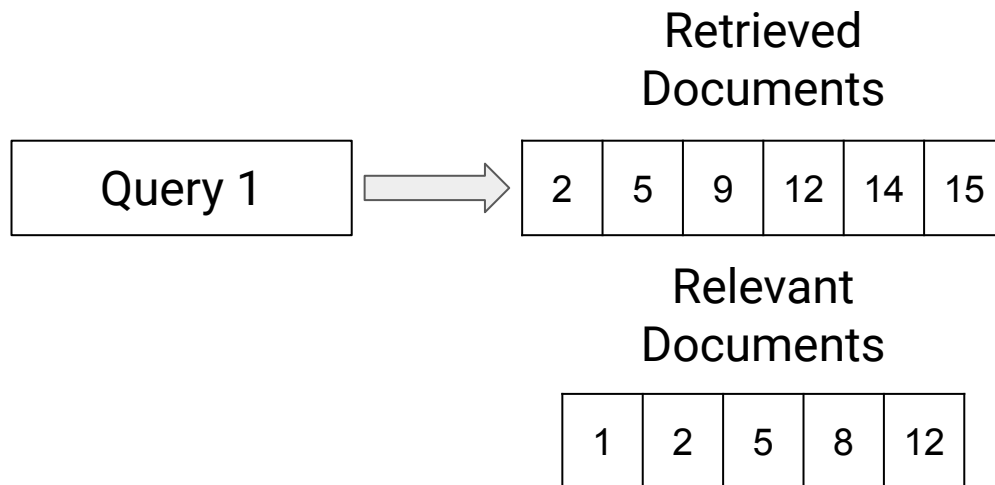
- Recall

How good are the retrieved documents?



- Recall: Fraction of relevant documents in the collection that are retrieved

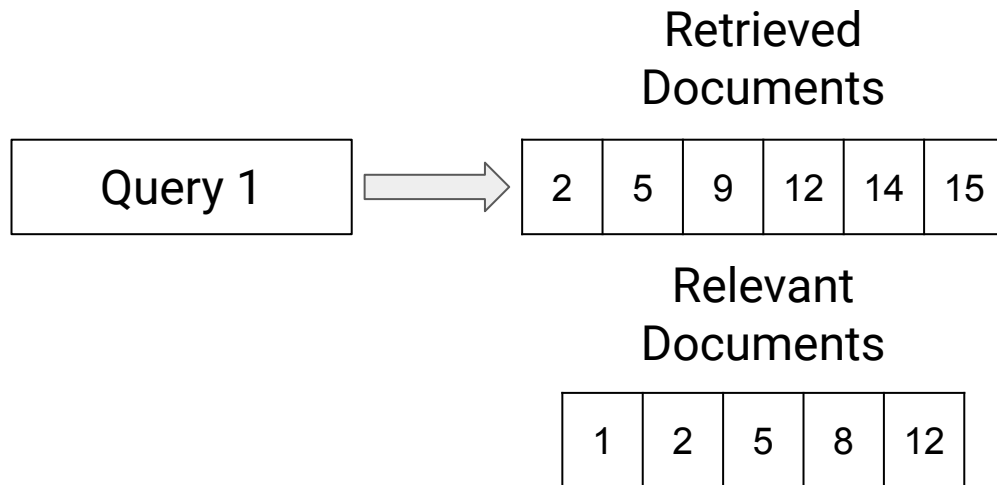
How good are the retrieved documents?



- Recall: Fraction of relevant documents in the collection that are retrieved

Recall =

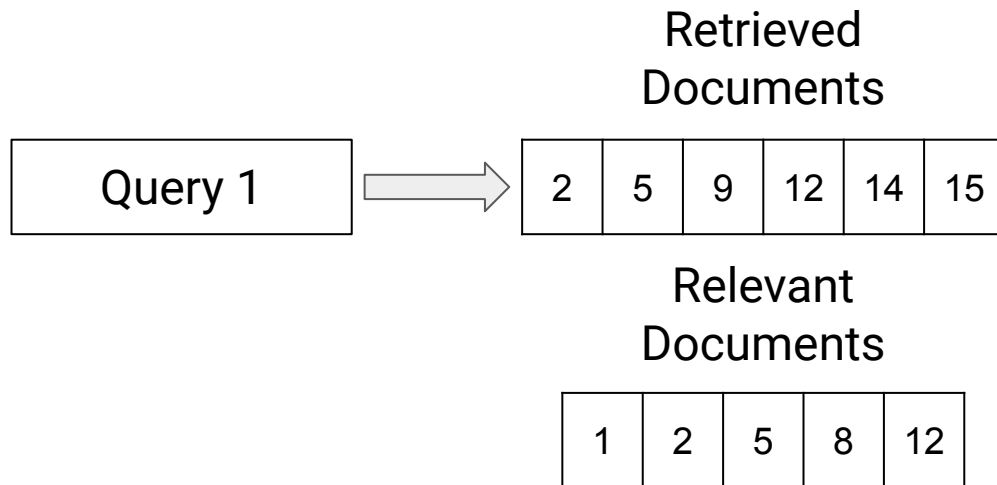
How good are the retrieved documents?



- Recall: Fraction of relevant documents in the collection that are retrieved

Recall = $\frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents in the collection}}$

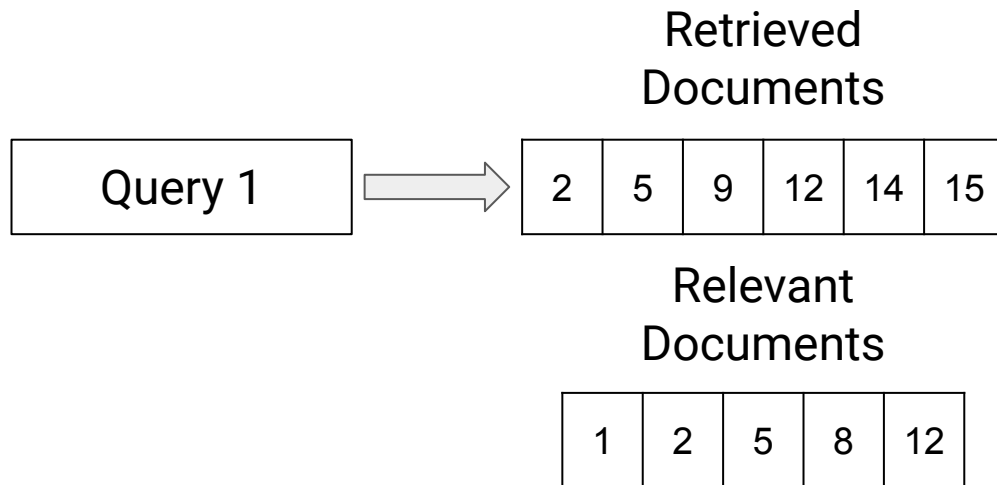
How good are the retrieved documents?



- Recall: Fraction of relevant documents in the collection that are retrieved

$$\text{Recall} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents}}$$

How good are the retrieved documents?



- Recall: Fraction of relevant documents in the collection that are retrieved

$$\text{Recall} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents}} = \frac{3}{5}$$